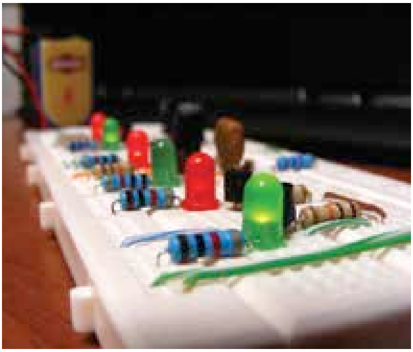
sonic sensor sends an ultrasonic wave (sound inaudible to humans). This ultrasonic wave reflects from obstacles directly in front of the sensor. The program on the Arduino™ board measures the time of reflection to calculate the distance using the simple formula distance = (speed x time)/2. Since the wave travels twice the distance due to reflection, we need to divide by '2' to get the actual distance. If the robot gets close to a wall while navigating (distance from wall directly in front is less than some value, say, 30 centimetres) rotate the ultrasonic sensor assembly to check distances on either sides to the robot. Steer the robot towards the direction where distance of the robot from the wall is the farthest.

This method is also used in the Micromouse event, where the distance of robots is measured from the wall. Then, an algorithm like flood fill helps the mouse travel to the centre and then calculates the route that the robot can travel in minimum time incase multiple routes exist.

You can check out this guide for detailed instructions and some code *http://dgit.in/JamesRobot*.



Prototyping of electronics is one of the first steps towards building anything tangible.

Prototyping

If all this sounds Greek and Latin to you, that's because it is. The word prototype is derived from the greek words for 'primitive form', and have connections to the greek word for 'first'.

Prototypes are one of the earliest model for a product you are building. Whenever you're making anything, you won't be arriving at the finished product at the first try. You will first need to model the product virtually, to see the design, then make a physical model of the product. When you make the first hardware model, you will find that not everything turns out as you wanted it to. There will be imperfections in the design and the build of the product and it has to be refined several times. Prototyping is involved in every step, right from the formulation, evaluation, to the execution of an idea. Prototypes can be a proof of concept, a form study, a model to get user feedback or it can also be used as a functional model for testing.